

Assignment-4: Multi-Modal Human Detection System

(CLO-3)

BSSE23 & BSCE22

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Human Computer Interaction

SE305T & MD445T (Spring-26)

Group Assessment

Information Technology University (ITU) – Lahore

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Total Marks = 100

1. Assignment Overview

This assignment asks your group to design and implement an interactive real-time detection system using a webcam. The system must present the user with a menu allowing them to choose one of four detection modes:

- **Lips Detection:** smile detection and mouth aspect ratio tracking [20 marks]
- **Eyes Detection:** blink counting and drowsiness alerting [20 marks]
- **Face Detection:** emotion recognition and head pose estimation [20 marks]
- **Hand Detection:** finger counting, gesture classification, and a gesture game [20 marks]

Input Types [20 marks]

Your system must allow the user to select the input source from the following options:

1. Live Webcam Feed

- Default real-time detection mode
- Continuous frame processing
- Suitable for interactive tasks (e.g., gesture game, blink detection)

2. Video File Upload

- User uploads a pre-recorded video
- System processes frame-by-frame
- Must support play, pause, and replay controls

3. Image Upload

- User uploads a static image
- System performs detection and displays annotated results
- Should clearly visualize detected features (e.g., landmarks, bounding boxes)

This assignment assesses your ability to apply computer vision techniques within an HCI context, with emphasis on usability, accuracy, and design.

2. Background & Context

Human Computer Interaction is increasingly driven by natural modalities gaze, expression, gesture, and posture rather than traditional mouse-and-keyboard input. Computer vision libraries such as OpenCV, MediaPipe, and DeepFace have made it feasible to deploy such systems on a standard laptop with a built-in webcam.

Your task is grounded in this reality. You will use these libraries to build a prototype that could serve as the front-end interaction layer for a range of real applications: driver monitoring, educational engagement tracking, pain assessment, and hands-free control systems.

3. Required Libraries & Setup

Install the following Python libraries before starting:

Library	Install Command	Purpose
opencv-python	pip install opencv-python	Webcam capture, drawing, Haar cascade
mediapipe	pip install mediapipe	Face mesh (468 pts), hand landmarks
deepface	pip install deepface	Emotion recognition from face region
numpy	pip install numpy	Numerical operations on landmarks

All libraries run on Windows, macOS, and Linux. A standard built-in webcam is sufficient. No GPU is required, though a dedicated GPU will improve DeepFace inference speed.

4. Technical Requirements

Your program must fulfil all of the following requirements. Each module contains three graded tasks (T1, T2, T3) of increasing difficulty.

4.1 Module 1: Lips Detection

Use MediaPipe FaceMesh (468 landmarks) to track the lips in real time.

Task	Name	Description & Acceptance Criteria
T1	Smile Detection	Detect whether the user is smiling using lip-corner y-positions relative to the mouth centre. Display 'Smiling' or 'Neutral' label on the frame in real time. The label must update within one second of the expression change.
T2	MAR Tracking	Compute and display the Mouth Aspect Ratio (MAR) on every frame. Highlight the video frame border in RED when MAR exceeds 0.55 (open mouth / yawn). Display the current MAR value to two decimal places.
T3	Lip-Sync Counter	Count the number of mouth open-close cycles (each OPEN then CLOSE transition = 1 event). Display the running count on the frame. Use hysteresis (two distinct MAR thresholds: open >

		0.55, close < 0.35) to avoid false triggers.
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4.2 Module 2: Eyes Detection

Use MediaPipe FaceMesh to extract eye landmarks and compute the Eye Aspect Ratio (EAR).

Task	Name	Description & Acceptance Criteria
T1	Blink Counter	Count blinks by detecting EAR transitions below 0.25 (closed) and back above 0.25 (open). Each complete close-open cycle = 1 blink. Display the total blink count on the frame.
T2	EAR Display	Display the real-time EAR value for both the left eye and right eye on the frame, updated every frame. Values should be shown to three decimal places.
T3	Drowsiness Alert	If the average EAR remains below 0.25 for more than 20 consecutive frames, overlay a prominent RED banner reading 'DROWSY! Wake Up!' across the frame. The alert must clear automatically once EAR returns above 0.25 for 5 consecutive frames.

4.3 Module 3: Face Detection

Combine OpenCV Haar cascade (bounding box), MediaPipe FaceMesh (head pose), and DeepFace (emotion).

Task	Name	Description & Acceptance Criteria
T1	Emotion Recognition	Use DeepFace.analyze() to classify the dominant emotion from the detected face region. Display the emotion label and its confidence percentage on the frame. Run inference every 5 frames to maintain smooth video.
T2	Head Pose	Estimate head orientation using MediaPipe landmark geometry. Compare nose-tip position to eye-midpoint and classify the direction as: Left, Right, Up, Down, or Forward. Display the direction label on the frame.
T3	Emotion History Log	Maintain a rolling 5-second buffer of detected emotions with timestamps. Display the most frequently detected emotion in that 5-second window as a 'Recent Mood' label on the frame.

4.4 Module 4: Hand Detection

Use MediaPipe Hands to detect up to two hands and extract 21 landmarks per hand.

Task	Name	Description & Acceptance Criteria
T1	Finger Counter	Count the number of extended fingers on each detected hand using tip vs PIP joint comparison. Handle the thumb separately with x-axis comparison. Display the count next to each hand.
T2	Gesture Label	Map finger count configurations to gesture labels: Fist, One, Peace, Three, Four, Open Hand, High Five, Thumbs Up. Display the gesture name and an emoji on the frame.

T3	Gesture Game	Display a random target gesture on screen. Award 1 point when the user holds the matching gesture for 1 continuous second. Then show a new random target. Display the score and a progress timer bar.
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5. Code Structure Requirements

Your submitted code must follow these structural rules:

- Single Python file named: `hci_detection_YourGroupmembername.py`
- A console menu lets the user choose input mode (1-3) and the detection mode (1-4) at runtime
- ESC key returns the user from any detection mode to the main menu
- Each module is implemented in its own clearly named function
- All functions must include a docstring explaining parameters and return values
- No global mutable state — pass data as function parameters or return values
- Code must run without modification on a standard laptop with a working webcam

Submission Details:

- Deadline: Sunday, May 3, 2026, at 11:59 PM.
- Marvel prototype link (view + interactive testing)
- PDF report
 - A table showing which member implemented which tasks in %.
 - Each member must write a short paragraph (min. 150 words) describing their personal contribution
- Include: All group members name, student ID, and the project title in the document.
- Ensure clarity, coherence, and proper formatting in your responses.